

Dissecting Atomic Facts

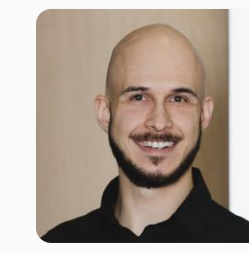
Visual Analytics for Improving Fact Annotations in Language Model Evaluation



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Motivation

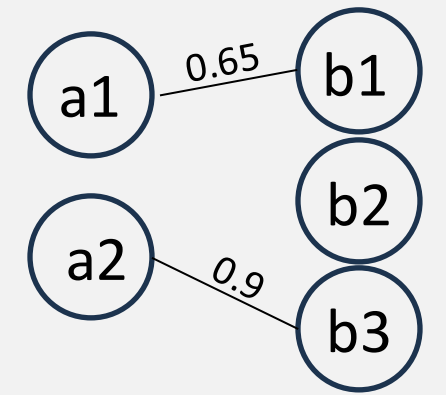
- LLMs are prone to making up false statements
→ Hallucinations
- Tools are needed to evaluate LLM outputs for factual correctness
- Meaningful LLM Evaluation paves the way for reliable AI in sensitive domains

Atomic Facts

- Smallest indivisible factual statement
→ Minimal
- Self-contained statement expressing one relation or attribute
→ Independent
- What constitutes an atomic fact?

LLM Evaluation using Atomic Facts

- Methods like FactScore [2] evaluate the factuality of LLMs
- Inspired by Summary Content Units from Summarization Evaluation [1]
- Utilize atomic facts
- How to match facts? Similarity based matching
- Precision $f(y) = \frac{1}{|A_y|} \sum_{a \in A_y} 1[a \text{ is supported by } \mathcal{C}]$



Finding

- Difference in granularity of extracted atomic facts
→ Low inter-annotator agreement
- Annotators treat Dependency resolution, Conditionals and Conjunctions differently
- Ex.: The dog was brown **and** very tall – How many facts?

Text-View

Aggregate plain text into knowledge graph

Graph-View

Show extracted facts by the annotators

Facts-View

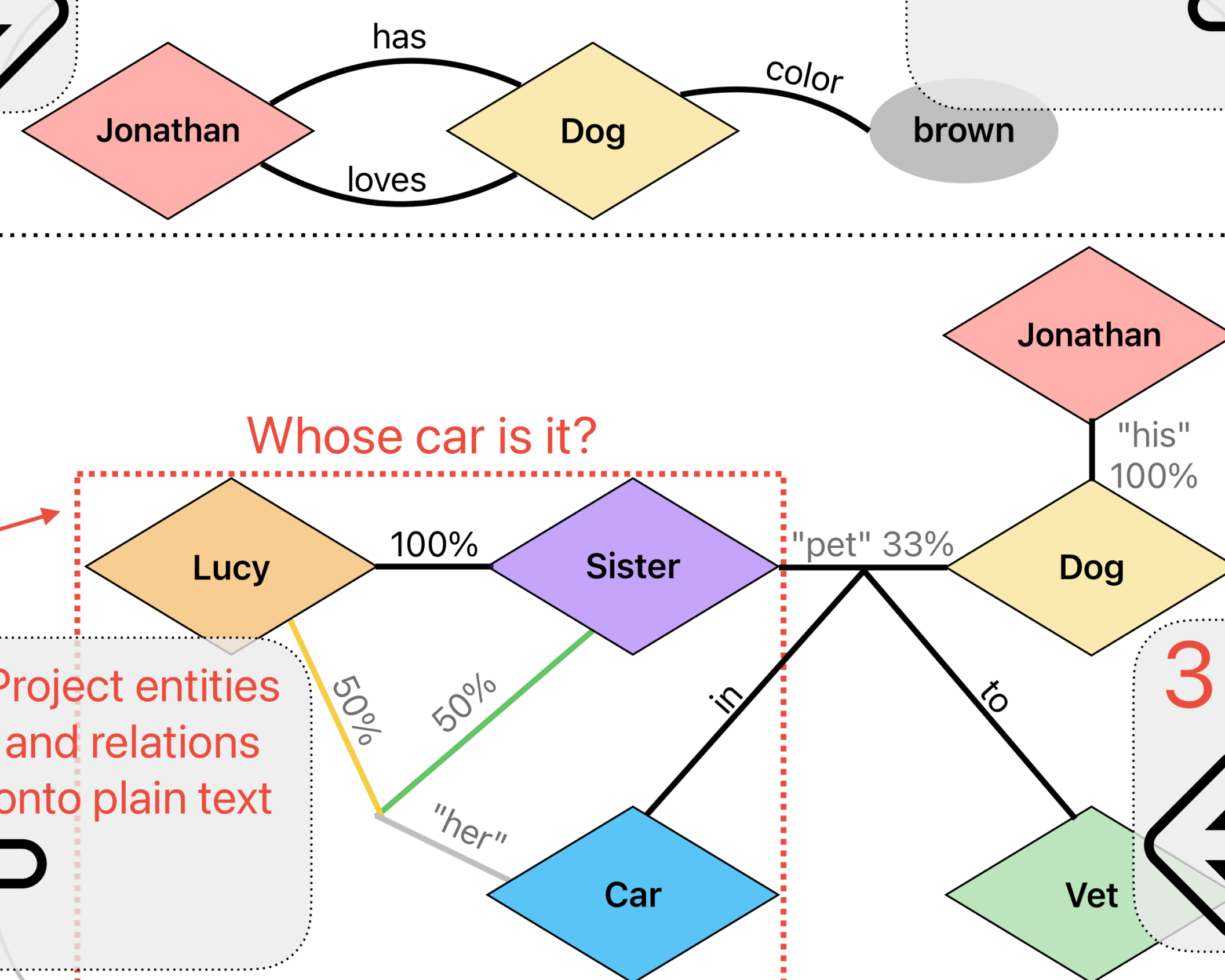
At work, Jonathan answered some emails. He attended a meeting. No one said anything unusual.

Jonathan has a brown dog, that he loves.

His colleague Lucy has two orange cats.

Her sister took his pet to the vet in her car.

At 5:00 PM, he turned off his computer and left the office to finally greet his dog at home.



Annotator 1, Human (expert):

- Jonathan has a brown dog
- Lucy has two orange cats
- Lucy's sister took Jonathan's dog to the vet in her car

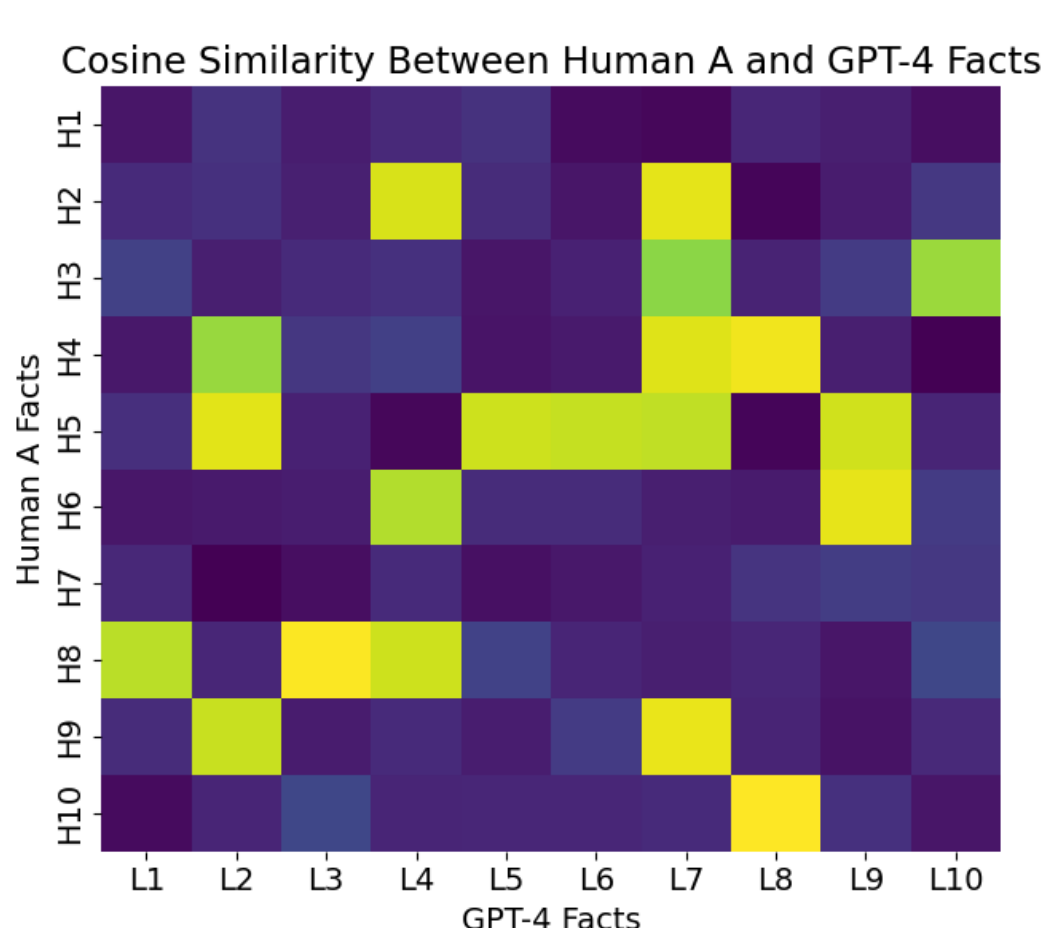
Annotator 2, Human (crowdworker):

- Jonathan has a dog
- There is a brown dog
- Lucy has two cats
- Lucy's cats are orange
- Lucy's sister has a car
- Lucy's sister took Jonathan's pet to the vet
- Lucy has a sister

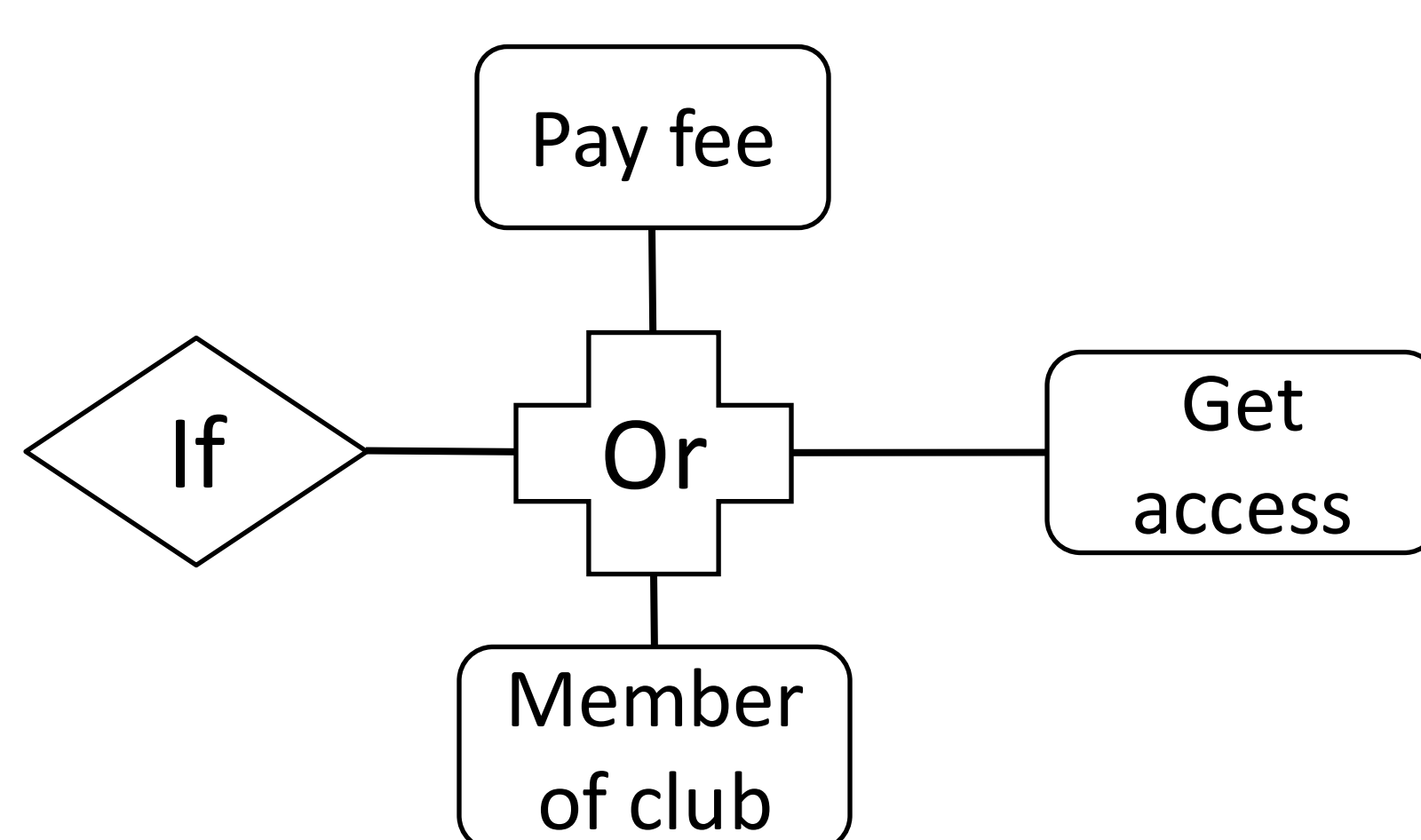
Aggregate extracted facts into KG

- Jonathan has a brown dog
- Lucy has two cats
- Her sister took his pet to the vet in Lucy's car

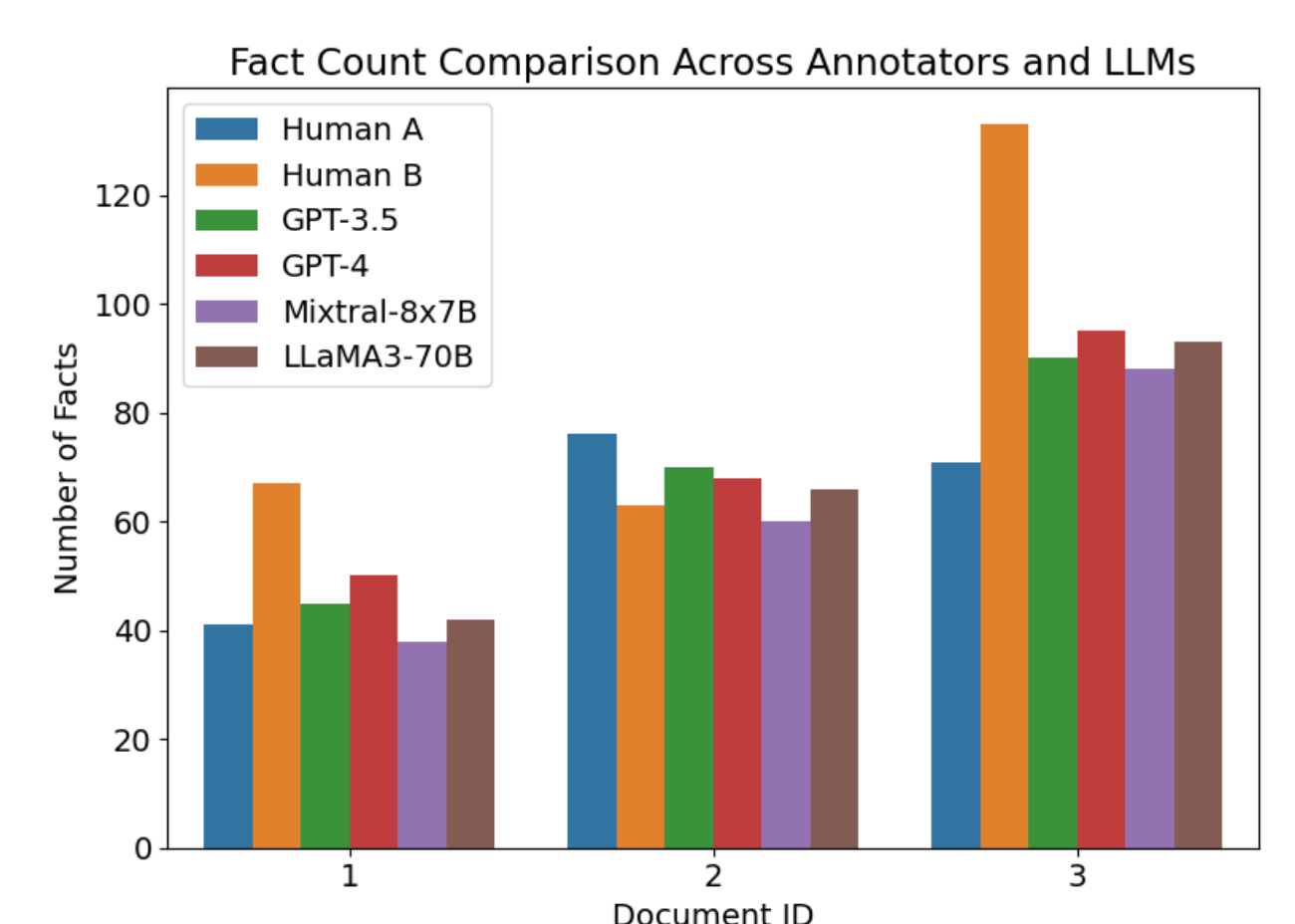
- Semantic highlighting of extracted facts
- Interactive selection highlights parts in the Graph-View



- Automatic knowledge graph (KG) generation
- Create KGs from original source text and extracted facts
- Comparison through small multiples



- Shows facts extracted by human and LLM annotators
- Entity highlighting
- Ambiguity highlighting



Similarity between Atomic Facts
→ Fact Matching

Visualize Conditionals
with Tree-Viewer

Fact Count Histogram
→ Annotator Agreement

Outlook

- Visual Analytics Application under development
- Features such as semantic clustering help in finding ambiguities
- Contradiction and redundancy detection help to spot disagreements
- Once agreement converges: **Human-in-the-loop LLM-guided annotation**

Conclusion

- A Visual Analytics based workflow is necessary to address the problems
- Analytic features like **detection of annotation inconsistencies** are required
- Iteratively **revise annotation guidelines**
- Goal: **Converge inter-annotator-agreement**

[1] Nenkova and R. J. Passonneau. Evaluating content selection in summarization: The pyramid method. In Hum. Lang. Technol. Conf. North Am. Chapter Assoc. Comput. Linguist., pp. 145–152, 2004
[2] Min, K. Krishna, X. Lyu, M. Lewis, W. Yih, P. W. Koh, M. Iyyer, L. Zettlemoyer, and H. Hajishirzi. FactScore: Fine-grained atomic evaluation of factual precision in long form text generation. In Proc. 2023 Conf. Empir. Methods Nat. Lang. Process., pp. 12076–12100, 2023. doi: 10.18653/V1/2023.EMNLP-MAI

